

Dynamic Programming: Further Examples

Design and Analysis of Algorithms

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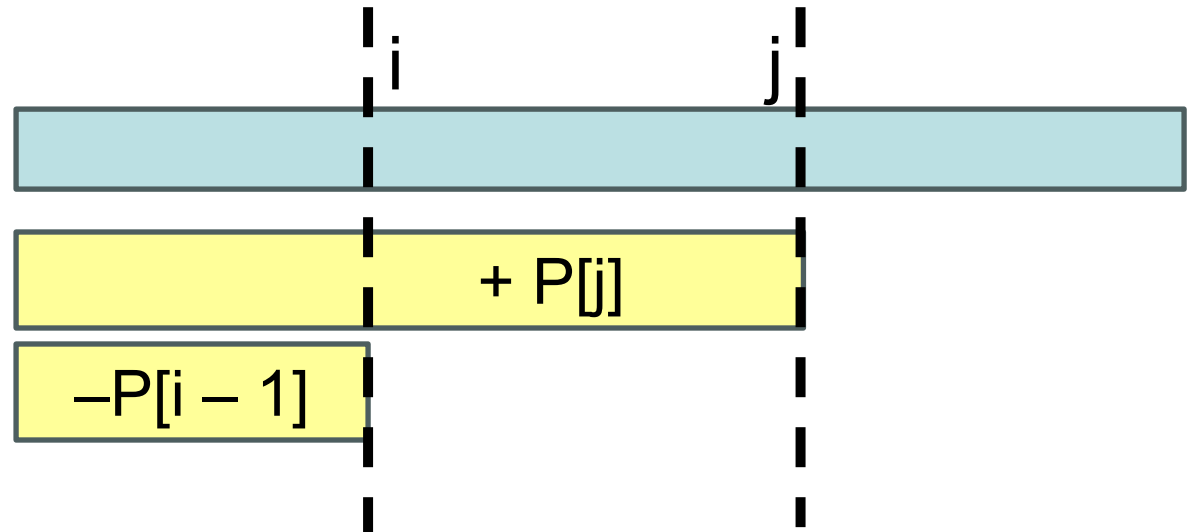
Warm-Up: Maximum-Value Subarray

- Given an array $A[1..n]$, find a contiguous subarray $A[i..j]$ that has maximum sum.
(Note: boring if all $A[i]$'s nonnegative)



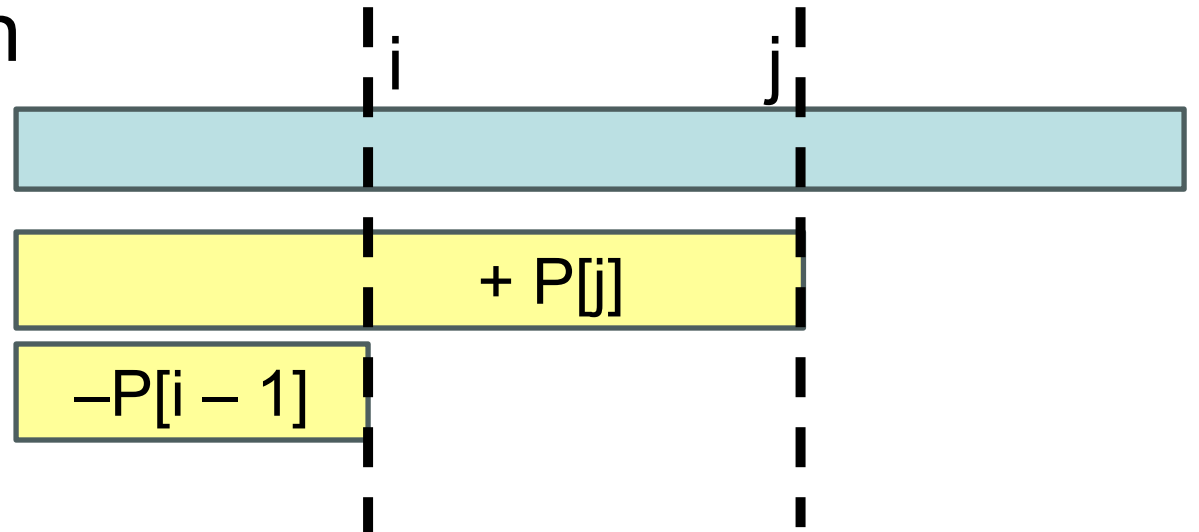
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- Easy to extend to find max sum over all subarrays **with some minimum length L** .



Aside: Binary Search on Answer and Ratio Objectives

- Given an array $A[1..n]$, find a contiguous subarray $A[i..j]$ that has maximum average.
- Optimal solution trivial! Just take largest array element.
- Ok, what about the subarray of length at least L having the largest average?
- Recall: simple $O(n)$ answer for sum instead of average.



Binary Search on Answer

- Often easier to check if the optimal solution of a problem is $\geq X$ than to compute it directly.
- If so, just binary search on X !
- In our case, can we easily test if there exists a subarray $A[i..j]$ of length at least L for which the average is $\geq X$?

$$\Leftrightarrow \text{for which } \frac{A[i] + \dots + A[j]}{1 + \dots + 1} \geq X$$

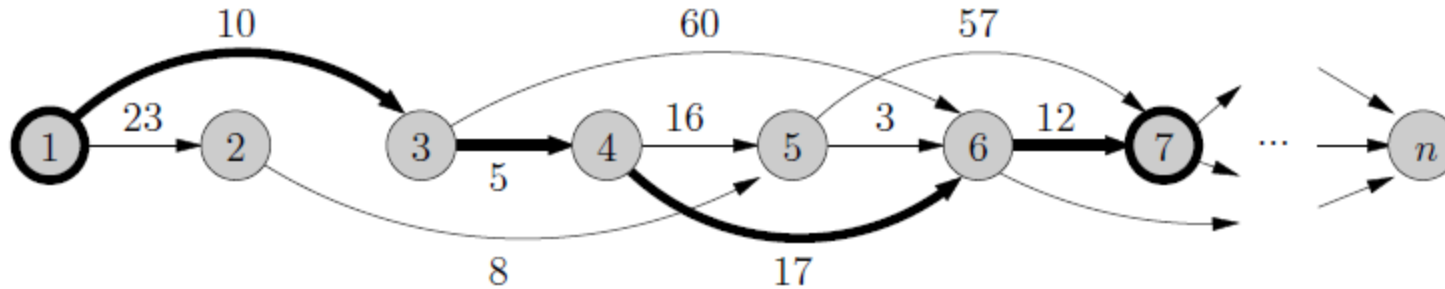


$$\Leftrightarrow \text{for which } A[i] + \dots + A[j] \geq X + \dots + X$$

$$\Leftrightarrow \text{for which } (A[i] - X) + \dots + (A[j] - X) \geq 0$$

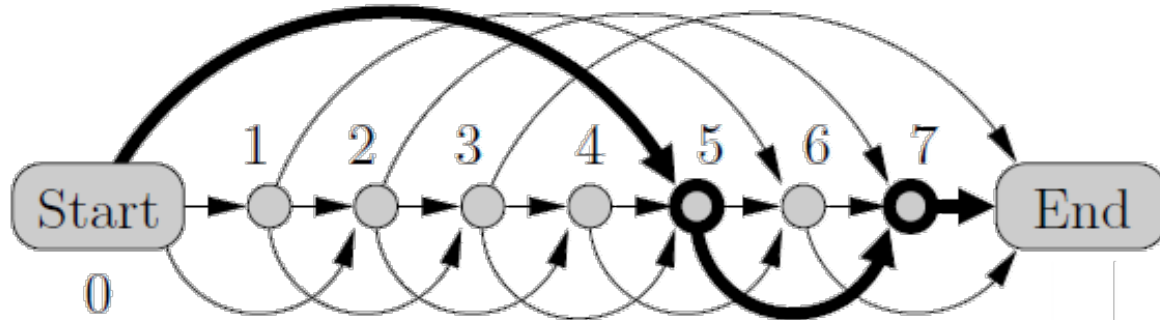
Acyclic Shortest Paths

- Find the shortest path from a source node to all other nodes in a Directed Acyclic Graph (DAG).

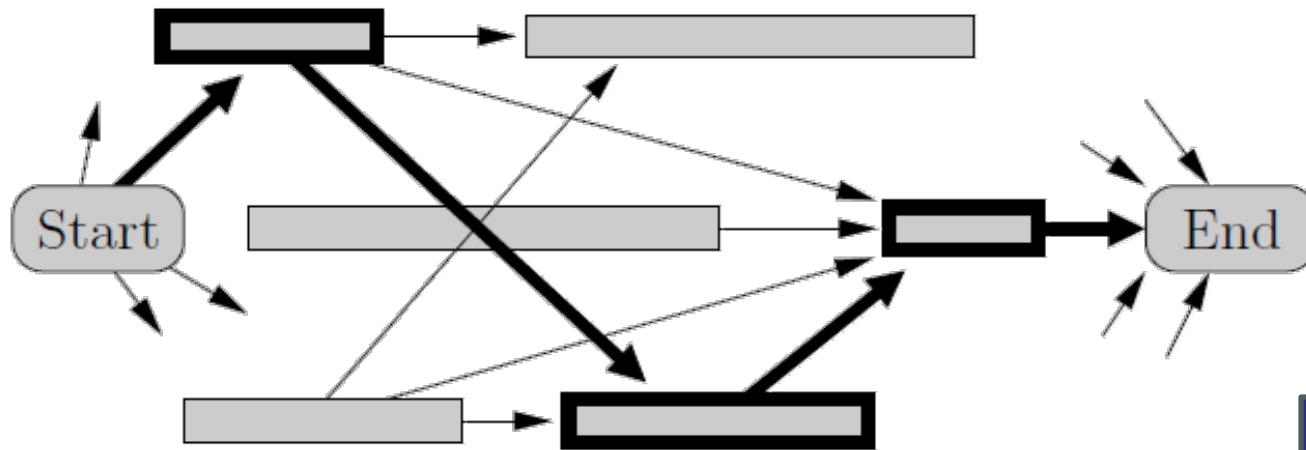


- As we'll see in the next module any DAG can be “topologically sorted” in linear time so all its edges point from left to right (a common preprocessing step).
- This problem is somewhat generic, in that many DP algorithms can be viewed as calculations on DAGs.

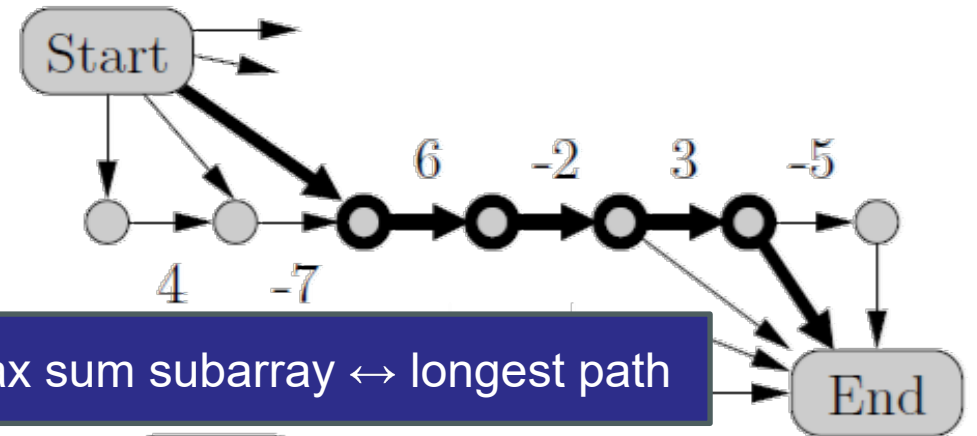
DP Algorithms as Calculations in DAGs



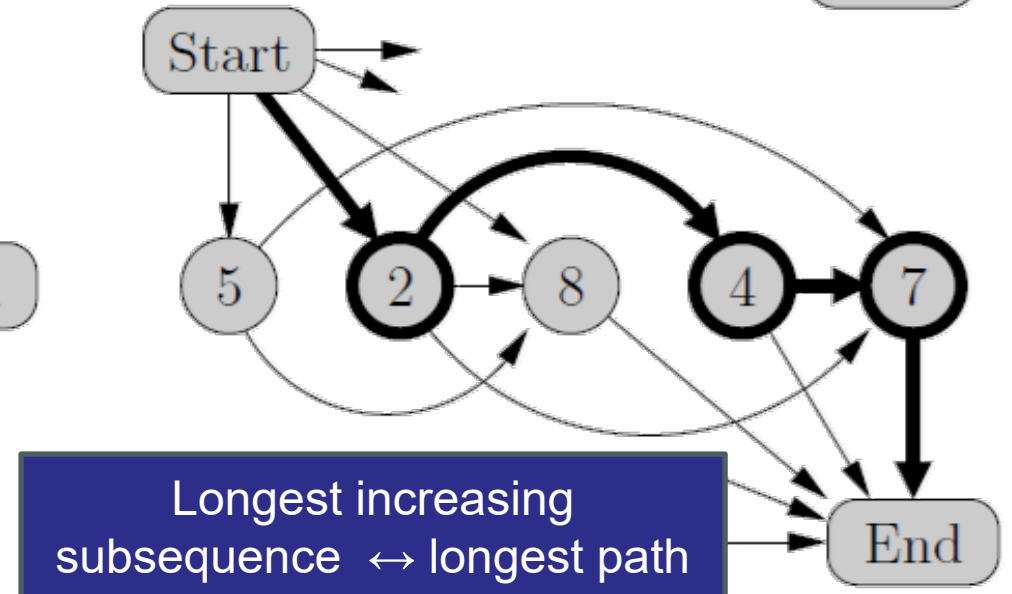
Making 8 cents in change with a shortest path representing coins of denominations 5, 2, 1



Maximum length set of intervals as a longest path problem (edge weights == interval lengths)



Max sum subarray $\leftrightarrow$ longest path



Longest increasing subsequence $\leftrightarrow$ longest path